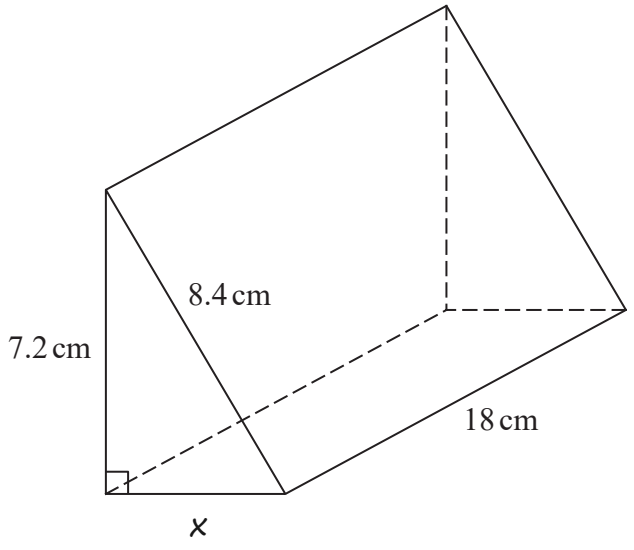
26 Here is a triangular prism.



Work out the volume of the prism.
Give your answer correct to 3 significant figures.

The 8.4 cm side is the hypotenuse.

$x^2 = 8.4^2 - 7.2^2 = 18.72$

1 mark
Use Pythagoras

$x = \sqrt{18.72} = 4.32666153...$ cm

1 mark
Take the square root

Triangle area = ½ × base × height

$= \frac{1}{2} \times \sqrt{18.72} \times 7.2$ cm^2

1 mark
Find the triangular area

Volume = triangle area × prism length

$= \frac{1}{2} \times \sqrt{18.72} \times 7.2 \times 18$

1 mark
Multiply by the length 18

$= 280.367667...$ cm^3

Volume = 280 cm^3 (3 significant figures).

1 mark
Keep full calculator precision until final rounding

280 cm^3

(Total for Question 26 is 5 marks)